Description:

The Larper-2 is a J-class rocket motor that uses casted KNSB propellant in a BATES grain configuration.

Technical Data:

All data shown is theoretical/simulated values.

Motor Designation: J624 (67%)

Total Impulse: 1068.15 Ns

Average Thrust: 646.5 N

Delivered Isp: 133.32 s

Burn Time: 1.68 s

Average Pressure: 837.42 psi

Peak Pressure: 931.53 psi

Propellant Mass: 817.02 g

Port to Throat Ratio: 4.25

Initial Kn: 366.29

Peak Kn: 411.25

Ideal Thrust Coefficient: 1.61

Delivered Thrust Coefficient: 1.48

Research Needs:

1. Static thrust characterization and experimental validation of the motor's predicted performance using an appropriate supervised facility and instrumentation.

2. Test an experimental liner and grain inhibitor material to validate the viability in rocket motors.

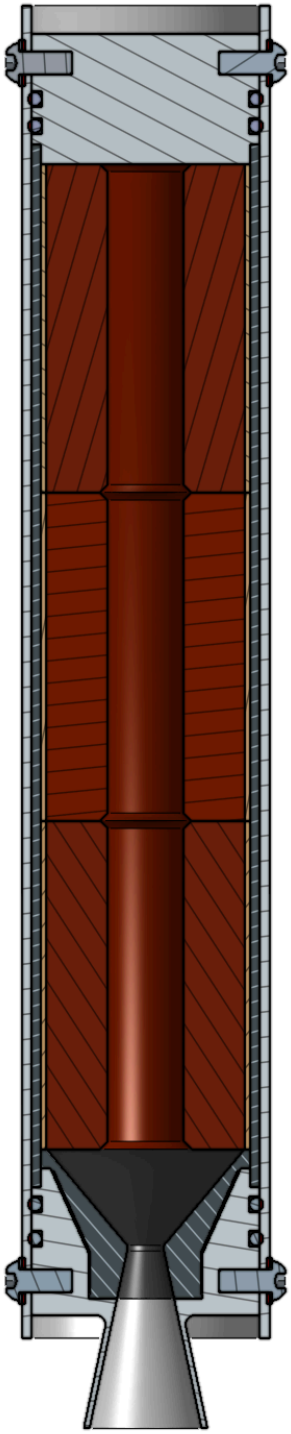


Fig 1. Larper-2 Rocket Motor Cross Section